

Education

H.B.Sc., Statistics, Astronomy & Astrophysics, Mathematics
University of Toronto 2018-2022

Research Experience

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| 2021 May - 2021 Aug | Undergraduate Summer Research Fellowship
<i>Canadian Institute for Theoretical Astrophysics</i>
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| 2020 Aug - 2021 May | Undergraduate Research Course
<i>Canadian Institute for Theoretical Astrophysics</i>
Supervised by Prof. Gwendolyn Eadie and Prof. Norman Murray . Study of mass distribution of the Milky Way in a Bayesian framework. Implemented a hierarchical Bayesian model for modelling the mass of galactic halos using an adaptive gradient-based sampling algorithm, increasing computational efficiency by two orders of magnitude over existing code. |
| 2020 May - 2020 Aug | Summer Undergraduate Research Program
<i>Dunlap Institute for Astronomy and Astrophysics, University of Toronto</i>
Supervised by Prof. Allison Man . Study of molecular gas in a strongly-lensed galaxy group at $z = 2.9$. Awarded best poster in student competition. Analyzed ALMA data to infer molecular gas masses in several high-redshift galaxies. Resulted in a first-author publication. |

Refereed Publications

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2. **Jeff Shen**, Gwendolyn M. Eadie, Norman Murray, Dennis Zaritsky, Joshua S. Speagle, et al. “Estimating the Mass Distribution of the Milky Way with Bayesian Multilevel Models, the No-U-Turn Sampler, and Halo Stars from the H3 Survey”, in prep.
 1. **Jeff Shen**, Allison W. S. Man, Johannes Zabl, Zhi-Yu Zhang, Mikkel Stockmann, Gabriel Brammer, Katherine E. Whitaker, and Johan Richard. “Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$ ”, accepted for publication in *ApJ*. <https://arxiv.org/abs/2105.11572>

Honours and Awards

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| 2021 May - 2021 Aug | Undergraduate Summer Research Fellowship, (\$8396 CAD)
Canadian Institute for Theoretical Astrophysics, University of Toronto |
| 2020 May - 2020 Aug | Summer Undergraduate Research Program Award (\$9,500 CAD)
Dunlap Institute for Astronomy and Astrophysics, University of Toronto |

Presentations

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| 2021 May | Stellar Stats Workshop, UofT/Online
Lightning talk: “Estimating the mass distribution of the Milky Way with Bayesian multilevel models.” |
| 2021 May | Canadian Astronomical Society (CASA) AGM, Online
Poster presentation and lightning talk: “Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$.” |
| 2021 Apr | Vancouver Area Cosmology Meeting, UBC/SFU/TRIUMF/Online
Presentation: “Molecular gas in gravitationally lensed galaxies at $z = 2.9$.” |
| 2020 Oct | REQUIEM Galaxies Telecon, Online
Presentation: “Molecular gas in gravitationally lensed galaxies at $z = 2.9$.” |

Media

2021 Jul **National Radio Astronomy Organization (NRAO) eNews**
Digital newsletter feature: “Molecular gas in high redshift galaxies”, **Jeff Shen** and Allison Man. <https://science.nrao.edu/enews/14.7>

Outreach

2019 - Present **Space Science Journalist (Current in Space segment)**, *The Star Spot* podcast
Curate, write and record stories about the latest research and news in astronomy.
Episodes 177 - present.

Technical Skills

Data: Stan, R, Python, NumPy, SciPy, Pandas, PyTorch, Scikit-Learn, Matplotlib, SQL
Other: Rust, BLAS/LAPACK, Astropy, L^AT_EX, regex, bash, vim, git, Docker, AWS, SymPy

Volunteering

2015 - 2018 **Python Workshop Lead**, Sir Winston Churchill Secondary, Vancouver, B.C.
Coordinated and delivered workshops to prepare students for national programming contest.